SkillGap AI

Job Skill Gap Analyzer

B.Tech CSE Mini Project Documentation

Project Type	Web Application
Domain	Career Guidance / Education Technology
Development Platform	Base44 AI
Target Users	B.Tech / CSE Students
Live Application	https://spicy-bridge-your-skill.base44.app

Prepared for academic mini-project submission and GitHub documentation.

1. About the Project

Students often learn programming languages, frameworks, tools, and other technologies without knowing whether those skills are relevant to the job they want.

SkillGap AI is a web application that helps students understand the skills required for a selected job role, compare those requirements with their current skills, identify skill gaps, and follow a personalized learning path.

The application is designed to bridge the gap between a student's current abilities and the skills expected for their target career.

2. Problem Statement

Students struggle to know exactly what skills they need for a specific job role.

Students frequently face questions such as:

• What skills are required for my target job?

• Which skills do I already have?

• Which skills am I missing?

• What should I learn first?

• Am I ready for an internship or placement?

• What projects should I build?

3. Proposed Solution

SkillGap AI provides a centralized platform where students can compare their current abilities with the expected skills for a selected career role.

The system follows the flow: Student Profile → Target Job Role → Current Skills → Skill Gap Analysis → Job Readiness → Learning Roadmap → Project Recommendations → Progress Tracking.

4. Objectives

• Help students understand job-specific skill requirements.

• Identify the difference between current skills and required skills.

• Provide a personalized learning roadmap.

• Recommend practical projects for skill development.

• Help students track their learning progress.

• Improve awareness of industry-relevant technologies.

• Provide a simple and user-friendly career preparation platform.

5. Key Features

Student Profile: Store educational details, current skills, programming languages, web technologies, databases, tools, soft skills, and preferred job roles.

Job Role Selection: Explore roles such as Python Developer, Frontend Developer, Backend Developer, Full Stack Developer, Data Analyst, Data Scientist, Machine Learning Engineer, Cybersecurity Analyst, Cloud Engineer, and DevOps Engineer.

Skill Gap Analysis: Categorize skills into Strong, Developing, and Missing.

Job Readiness Score: Present an easy-to-understand readiness indicator based on the student's skill profile.

Personalized Learning Roadmap: Provide a structured sequence of skills and learning stages.

Project Recommendations: Recommend practical projects based on missing skills.

Career Explorer: Show job descriptions, skills, technologies, difficulty, and career paths.

Skill Library: Organize skills into programming, web, databases, AI/ML, cloud, cybersecurity, DevOps, data, tools, and soft skills.

Progress Tracking: Track completed, learning, and remaining skills.

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Example Job Readiness Display: 78% | Strong Skills: 8 | Developing Skills: 4 | Missing Skills: 3
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6. How It Works

Step 1 – Create Profile: Enter educational information and current skills.

Step 2 – Select Target Career: Choose the desired job role.

Step 3 – Skill Analysis: Compare current skills with required skills.

Step 4 – Identify Skill Gaps: Find strong, developing, and missing skills.

Step 5 – Generate Roadmap: Create a structured learning path.

Step 6 – Build Projects: Use project recommendations to practice required skills.

Step 7 – Track Progress: Update learning progress and improve readiness.

7. Application Workflow

Student Login → Student Profile → Select Job Role → Enter Skills → Skill Gap Analysis → Missing/Strong Skills → Learning Roadmap → Project Recommendations → Progress Tracking

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Student Login
↓
Student Profile
↓
Select Job Role
↓
Enter Current Skills
↓
Skill Gap Analysis
↓
Learning Roadmap
↓
```

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Project Recommendations
↓
Progress Tracking
```

8. Technology Used

Base44 AI: Primary AI-assisted application development platform used to create the web application.

Frontend: Responsive web interface, modern UI components, dashboards, forms, charts, and interactive elements.

Backend: Base44 application services, authentication, application logic, and data management.

Database: Structured data for users, skills, job roles, skill requirements, projects, and learning progress.

Deployment: Base44 web hosting.

9. Project Architecture

Student → SkillGap AI Interface → Application Logic → Users / Skills / Job Roles → Skill Gap Analysis → Learning Path + Project Recommendations.

```
Student
↓
SkillGap AI Interface
↓
Application Logic
↓
Users • Skills • Job Roles
↓
Skill Gap Engine
↓
Learning Path + Project Recommendations
```

10. Target Users

• B.Tech students

• CSE students

• College students

• Students preparing for internships

• Students preparing for placements

• Beginners entering software development

• Students exploring technology careers

11. Advantages

For Students: Provides career direction, reduces learning confusion, identifies missing skills, encourages project-based learning, and supports progress tracking.

For Colleges: Can potentially support placement preparation, skill assessment, career guidance, internship preparation, and student development programs.

12. Future Enhancements

• Advanced AI career assistant

• Resume analysis and job-role matching

• Industry skill-trend tracking

• Job description analysis

• Gamification with badges and milestones

• GitHub integration

• Interview preparation and mock interviews

13. Limitations

Skill requirements can vary between companies.

Job readiness scores are indicative and are not official hiring assessments.

Recommendations depend on available job-role and skill data.

Industry requirements change over time.

The application does not guarantee employment.

14. Testing

• User registration and login/logout

• Profile creation and editing

• Job role selection

• Skill selection and analysis

• Job readiness display

• Learning roadmap

• Project recommendations

• Progress tracking

• Navigation

• Responsive design across screen sizes

15. Expected Outcome

The expected outcome is a user-friendly platform that helps students understand their career requirements and build a structured path toward their desired job role.

Instead of simply asking *“What should I learn?”*, students can determine what their target job requires, what they already know, what they are missing, and what they should learn next.

16. Mini Project Information

Project Title: SkillGap AI – Job Skill Gap Analyzer

Project Type: Web Application

Domain: Career Guidance / Education Technology

Target Users: B.Tech / CSE Students

Development Platform: Base44 AI

Deployment: Base44 Web Hosting

Status: Completed / Prototype

17. Live Application

Live Demo: https://spicy-bridge-your-skill.base44.app

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https://spicy-bridge-your-skill.base44.app
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18. Conclusion

SkillGap AI addresses a common challenge faced by students: understanding the difference between their current skills and the skills expected for a target job role.

By combining skill-gap analysis, personalized learning roadmaps, project recommendations, and progress tracking, the application provides a structured approach to career preparation.

The project demonstrates how AI-assisted development platforms such as Base44 can be used to rapidly create and deploy a practical web application for a real-world problem.

19. Author

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